Hugo HADJUR

PhD in Computer Science — Artificial Intelligence — FRANCE — hugo.hadjur@gmail.com

Academic Page   Google Scholar   LinkedIn

I have a **PhD** in informatics with **ENS de Lyon** with strong dominance in **AI**, and **NLP** experience.
Since 2023, I am working in applied **biostatistics and AI research** for **pharma and biotech**.
I **write peer-reviewed scientific papers, find and adapt state-of-the-art methods and collaborate with researchers outside my field**.

Work Experience

Since Dec. 2023
Permanent

AI & Statistical Methodologist at **Saryga** (France)

— **Tech projects**: semantic analysis using LLMs, text summarization, zero-shot text classification, paraphrase mining, development of R Shiny apps and R packages, geodata analysis, intern supervision

— **Biostatistics and Pharma projects**: translating state-of-the-art research into solution in biomarker research, survival analysis, quantitative decision-making methods and sample size optimization, lead the writing of peer-reviewed papers

Sept. 2020 - Aug. 2023
3-year contract

PhD Candidate and Assistant Professor at **École normale supérieure de Lyon** and **aivancity** (France)

PhD Thesis: **Designing sustainable, autonomous, and energy efficient Internet of Things systems, applied to smart beehives**
Advisorship of Laurent Lefèvre & Doreid Ammar, at LIP-Avalon laboratory

— **Teaching**: advanced data analysis and visualization, data preprocessing and visualization and coding for AI and data science (Python, Tableau)
— **Intern supervision**: connected beehives deployment & data analysis
— **Head of online course platform** aivancityX

Sep. 2019 - Aug. 2020 & Oct. 2018 - Dec. 2018
1-year contract

Instructor & Researcher at **emlyon Business School** (France)

— Research: Connected beehives, bees' data analysis (**IoT, machine learning, computer vision**)

— Teaching: **Python programming bootcamp** and **business analyst toolbelt** (SQL, Excel, Tableau)

Jan. 2019 - Sep. 2019
8-month contract

Data Analyst at **Schneider Electric**, Houston, TX (USA)

— Supported the business by building **reports/dashboards and KPIs** to monitor performance, conducted **database integration**
— **Communicated findings** and insights to cross-functional team members and management
— Tool used: **SQL, Tableau, Python**

Oct. 2017 - Sep. 2018
1-year exchange

Student Researcher at **Kyoto University's Ishii Lab** (Japan)

Master's thesis: master board games thanks to **Reinforcement Learning** (Q-Learning, SARSA, Deep Reinforcement Learning)

May-Sep. 2017
4-month internship

Data Science Research Internship at **CiNet** (Center for Information and Neural Networks, Osaka, Japan)

— **Conducted research** about the links between **Twitter data**, human behavior and brain data, resulting in a published peer-reviewed paper
— **Analyzed large data sets** ($\approx$ 500,000 Twitter users) thanks to **PCA, clustering & regression** techniques
— Worked in a **Japanese environment**

Publications

[under review] **Hugo Hadjur**, Julia Geronimi, Sandrine Guilleminot, Alessandra Serra, Libby Daniells, Marie-Karelle Riviere, Gaëlle Saint-Hilary, Pavel Mozgunov. **Evaluating methods for biomarker cut-off detection: a comparative study in small and unbalanced samples**, 2025-26

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Deep Reinforcement Learning for Energy-efficient Selection of Embedded Services at the Edge**, 2024 IEEE International Conferences on Internet of Things (iThings), Copenhagen, Denmark, 2024, pp. 67-74, 10.1109/iThings-GreenCom-CPSCom-SmartData-Cybermatics62450.2024.00034. hal-04708697

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Services Orchestration at the Edge and in the Cloud for Energy-Aware Precision Beekeeping Systems**, 2023 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), St. Petersburg, FL, USA, 2023, pp. 769-776, 10.1109/IPDPSW59300.2023.00129. hal-04091575

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Toward an intelligent and efficient beehive: A survey of precision beekeeping systems and services**, Computers and Electronics in Agriculture, Elsevier, 2022, 192, pp. 1-16. 10.1016/j.compag.2021.106604. hal-03483914v2

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Analysis of Energy Consumption in a Precision Beekeeping System**. IoT '20 - 10th International Conference on the Internet of Things, Oct 2020, Malmö, Sweden. pp. 1-11, 10.1145/3410992.3411010. hal-02973772

Kazuma Mori, **Hugo Hadjur**, Masahiko Haruno. **Natural Language Content Mediates the Association Between Active Interactions on Social Network Services and Subjective Well-Being**, Cyberpsychology, Behavior, and Social Networking, 2022, pp. 678-685. 10.1089/cyber.2021.0340

Alessandra Serra, Julia Geronimi, Sandrine Guilleminot, **Hugo Hadjur**, Marie-Karelle Riviere, Gaëlle Saint-Hilary, Pavel Mozgunov. **A Novel Approach to Assess the Predictiveness of a Continuous Biomarker in Early Phases of Drug Development**, Statistics in Medicine, 2025, 44: e70026. 10.1002/sim.70026

Communication

Posters

— **SnB** (Statistics and Biopharmacy Conference) - Paris, France, 2025. Evaluating methods for biomarker cutoff detection: a comparative study in small and unbalanced samples

— **ISCB** (Annual Conference of the International Society for Clinical Biostatistics) - Thessaloniki, Greece, 2024. Simple Approaches for Portfolio Quantitative Decision-Making

Presentations

— [accepted] **PSI 2026** (Statisticians in the Pharmaceutical Industry Conference) - Belfast, Northern Ireland, 2026. Evaluating methods for biomarker cut-off detection

— **GreenDays 2023** - Lyon, France, 2023. Optimize task selection under energy constraints in IoT systems thanks to reinforcement learning

— **Journées du GDR Réseaux et Systèmes Distribués** - Rennes, France, 2022. Designing connected, sustainable, autonomous and low-consumption systems for a distributed network of smart beehives

— **IoT'20** - Online, 2020. Analysis of Energy Consumption in a Precision Beekeeping System

Programming & Software

Programming:	**Python (PyTorch, Hugging Face Transformers, NumPy, Pandas, scikit-learn, Natural Language Toolkit NLTK), R, SQL**
Tech Experience:	**Large Language Models (LLMs), Transformers, Natural Language Processing (NLP), Reinforcement Learning (RL), Computer Vision, Diffusion Models, Model Optimization**
Design & Text:	**LaTeX, Adobe Photoshop, Microsoft Office, Inkscape**

Qualifications ✓ "Grande École" diploma + PhD

2020-2023 | PhD at **ENS de Lyon** (**LIP-Avalon** laboratory) & **aivancity**

Develop and model Internet of Things (IoT) systems and services based on Artificial Intelligence (AI) to support beekeepers in their work and preserve bees while studying the efficiency of connected beehives that are autonomous, sustainable, distributed and operate under limited energy budget.

2017-2018 | Exchange year at **Kyoto University's Graduate School of Informatics**

Alongside courses in English and Japanese, I worked toward my Master's thesis. This thesis proposes a reinforcement learning method applied to the game of bridge. The Q-Learning and SARSA techniques reach a level of play comparable to an intermediate human player.
— Classes in **English & Japanese** in the department of **social informatics**: **statistics, machine learning**

2015-2017 | French "Grandes Ecoles": **Ensimag, Grenoble Institute of Technology** (equivalent: Master's Degree)
— MMIS specialization: **programming, advanced statistics, data mining, database manipulation**

2013-2015 | Scientific preparatory classes at **CPP Grenoble** (equivalent: Bachelor's Degree)
— 2-year intensive university-level preparatory course to enter **highly competitive Ivy League engineering schools**

2007-2013 | Middle school and high school in Tokyo (4 years at **LFIT**) and in Singapore (1 year at **LFS**)
— Scientific baccalaureate with highest honours (French school)